

Introduction to Multimedia - ICT 1210

Tutorial 04.

2.

1. Differentiate mechanical waves and electromagnetic waves

Mechanical Waves :- Require a physical medium to propagate and can't travel through a vacuum.

Eg :- Sound waves, water waves

Electromagnetic Waves :- Do not require a physical medium and can travel through a vacuum at the speed of light.

Eg :- light waves, radio waves, x-rays.

2. Differentiate between Sound and Audio.

Sound :- The physical vibrations we hear from the environment. Physical kinetic energy produced by mechanical vibrations propagating through a medium.

Audio :- Audio is the real voice recording or electronic version of the voice.

The electronic or digital representation and handling of sound.

3.

3. Describe Nyquist Theorem with example.

Nyquist-Shannon sampling rate must be at least twice the maximum frequency of the signal to avoid aliasing.

Example :- If a sound has maximum frequency of 20 kHz, the minimum sampling rate should be 40 kHz.

4. Pulse Code Modulation (PCM)

A method of representing quantized sampled audio as digital data. Each sample is rounded to the nearest integer value.

Producing quantized sampled output for audio is simply called Pulse Code Modulation.

5. File Size calculation.

File size = Sample rate $\times$ Bit resolution $\times$ time length $\times$ channels

Sample Rate	Bit value	Mono (Bytes)	Stereo (Bytes)
44.1 kHz	16	827,250	1,654,500
44.1 kHz	8	441,000	882,000
22.05 kHz	16	441,000	882,000
22.05 kHz	8	220,500	441,000
11.025 kHz	8	110,250	220,500

6. Differentiate MIDI and Digital Audio.

Digital audio :- Stores actual sound waveform; requires more storage

realistic playback, MIDI sends instructions to instruments. Digital audio stores waveform data.

MIDI :- Store instructions (notes, volume); Device dependent, requires synthesizer

MIDI sends instructions to instruments.

7. Explain Dolby sound system and DTS sound system.

Dolby

DTS

(Digital Theater system).

* Surround sound formats like Dolby Digital®, Dolby Pro Logic®

* Wildly used in cinema and home theaters.

*

* Competing surround sound technology.

* offering high quality multi-channel audio.

* DTS has created a range of

DTS Digital Theater Sound Formats.

8. Components of a sound card and functionalities.

* Sound card :- Internal / External expansion card for audio input / output.

* Functions :- Play-back, digitizing, recording, compressing.

(Sound card provides input and output of audio signals.

Main functions :- play back, digitizing, recording, compressing).

9. Explain 2 different Microphone types.

Dynamic Microphone :- Durable.

uses diaphragm and coil

Good for live sound

Condenser Microphone :- Sensitive

Requires power (phantom power)

used in studios for high-quality recording.